

# Quang Truong

✉ 251minhquang@gmail.com | 🏠 quang-truong.github.io | 📧 quang-truong | 💻 quang-cm-truong | 🎓 Google Scholar

## Professional Summary

Doctoral student with 5+ years of experience in software engineering, machine learning, and artificial intelligence. Skilled in deploying scalable ML systems and deep learning solutions. Expertise in **Graph Machine Learning**, **Relational Deep Learning**, **Time-series Modeling**, and **Agentic AI**, with publications in **ICML**, **NeurIPS**, **AAAI**, and **JMLR**. Strong background in industry research, large-scale model development, and open-source contributions. Experienced across the ML lifecycle, including data processing, model design, training optimization, and distributed computing.

## Work Experience

### Snap Inc.

Bellevue, WA

Research Intern (Mentor: **Neil Shah**)

Jun. 2026 - Present

- Lead research projects in the user modeling and personalization domains (sub-areas include graph modeling, generative recommendation, personalization, and ML efficiency).
- Build scalable research prototypes and evaluate them in large-scale machine learning scenarios.

### Nokia Corporation

Sunnyvale, CA

AI R&D Engineer Co-op (Mentors: **Liang Wu**, **Liangjie Hong**)

Jan. 2026 - May. 2026

- Designed, developed, and deployed advanced AI/ML models and algorithms to analyze and interpret complex data.
- Designed and implemented machine learning models to improve a wide range of applications, including search, forecasting, text mining, and more.
- Developed and implemented agentic-based systems for a wide range of applications, including anomaly detection, root-cause analysis, and more.

## Education

### Michigan State University

East Lansing, MI

Ph.D. in Computer Science

Aug. 2024 - May 2028

### Dartmouth College

Hanover, NH

M.S. in Engineering Sciences

Sep. 2022 - Jun. 2024

## Selected Publications

‡ indicates equal contribution.

### Peer-reviewed Publications

- [1] Plain Transformers are Surprisingly Powerful Link Predictors  
**Quang Truong**, Yu Song, Donald Loveland, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2026**  
*ICML'26: The 43rd International Conference on Machine Learning*.
- [2] A Pre-training Framework for Relational Data with Information-theoretic Principles  
**Quang Truong**, Zhikai Chen, Mingxuan Ju, Tong Zhao, Neil Shah, and Jiliang Tang. **2025**  
*NeurIPS'25: The 39th Annual Conference on Neural Information Processing Systems*.
- [3] TopoX: A Suite of Python Packages for Machine Learning on Topological Domains  
Mustafa Hajji<sup>‡</sup>, Mathilde Papillon<sup>‡</sup>, Florian Frantzen<sup>‡</sup>, ... **Quang Truong**, and ... Nina Miolane. **2024a**  
*The Journal of Machine Learning Research (Volume 25)*.
- [4] Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes  
**Quang Truong** and Peter Chin. **2024b**  
*AAAI'24: The 38th Annual AAAI Conference on Artificial Intelligence*.
- [5] Not All Data Matters: An Efficient Approach to Multi-Domain Learning in Vehicle Re-Identification  
**Quang Truong** and Bo Mei. **2021**  
*ITSC'21: The 24th IEEE International Conference on Intelligent Transportation Systems*.

## Open-source Projects

TopoX

- Provides reliable and user-friendly building blocks for computing and machine learning on topological domains that extend graphs.
- Adopted open-source software engineering best practices with comprehensive testing, documentation, and CI/CD pipelines.
- Starred 500+ times across 3 different libraries on GitHub.
- Published at the Journal of Machine Learning Research (JMLR).

Research Experience

DSE Lab, Michigan State University

Doctoral Research Assistant (Advisor: **Jiliang Tang**)

- Develop scalable transformers and characterize necessary conditions for the models to be strong Link Predictors (**Snap Research Collaboration**).
- Design and implement distributed training pipelines for large-scale graphs, optimizing performance across multiple GPUs and ensuring scalability for enterprise-scale data.
- Create automated testing frameworks and continuous integration pipelines to ensure code quality and system reliability across research projects.
- Develop specialized pre-training procedures for graph machine learning on relational databases, thereby enhancing predictive modeling for node-level classification, link prediction, and temporal forecasting tasks (**Snap Research Collaboration**).

East Lansing, MI

Aug. 2024 - Present

LISP Lab, Dartmouth College

Graduate Research Assistant (Advisor: **Peter Chin**)

- Introduced a novel approach to Topological Deep Learning (TDL) leveraging simple paths, significantly enhancing graph expressivity and surpassing the theoretical limitations of the 3-WL test.
- Developed Path Isomorphism Network, a state-of-the-art TDL model, which delivered outstanding performance on benchmark datasets: PROTEINS (78.8%), NCI1 (85.1%), NCI109 (84.0%), and IMDB-B (76.6%).
- Developed an anomaly detection pipeline for human behavior analysis using geospatially-tagged data, as part of the **IARPA-HAYSTAC** funded project.

Hanover, NH

Sep. 2022 - Jun. 2024

Vision Lab, University of Illinois at Chicago

Research Intern (Advisor: **Wei Tang**)

- Designed a context-aware 3D object detection architecture that captures both object-wise and object-context relationships, improving spatial understanding.
- Boosted model performance through cross-modality alignment, improving the feature representation of objects.

Remote

May 2021 - Aug. 2021

ML Lab, Texas Christian University

Undergraduate Research Assistant (Advisor: **Bo Mei**)

- Developed an advanced Vehicle Re-identification pipeline utilizing Generative Adversarial Networks (GANs) for adaptive domain learning, resulting in a 10% improvement in mean Average Precision (mAP) over the baseline.
- Engineered a novel filtering algorithm that reduces the training data requirement by 25% while maintaining model performance, significantly improving computational efficiency.

Fort Worth, TX

May 2019 - May 2021

Proposal Writing

- On the Conditions for Transformers to be a Strong Link Predictor

2025

PIs: Hui Liu, Jiliang Tang

Role: Drafted part of the proposal

Result: ICML'26 publication and awarded research grant.

Skills

|                         |                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming             | Python (Expert), C/C++ (Advanced), Java (Proficient), R (Proficient), MATLAB (Proficient)                                                                                 |
| Machine Learning        | Graph Machine Learning, Relational Deep Learning, Knowledge Graphs, Recommender System, Large Language Models, Computer Vision, Data Mining                               |
| ML Frameworks           | Pytorch, Pytorch Geometric, Pytorch Frame, NetworkX, TopoX, Pytorch Lightning, Hugging Face, Scikit-learn, Matplotlib, Numpy, Pandas                                      |
| DevOps & Infrastructure | WandB, Docker, Singularity, Pytest, Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, CI/CD, Monitoring, Distributed Systems                            |
| Software Engineering    | Spring, Agile, JIRA, SCRUM, Git, Linux/Unix, MongoDB, DuckDB, MySQL, Unit Testing, Performance Optimization, Automation, Algorithm Design, Data Structures, System Design |

**Professional Services** \_\_\_\_\_

**Conference Reviewer**

- Conference on Neural Information Processing Systems (NeurIPS) **2025, 2026**
- International Conference on Machine Learning (ICML) **2026**
- International Conference on Learning Representations (ICLR) **2024**
- Annual AAAI Conference on Artificial Intelligence (AAAI) **2025**
- Learning on Graphs Conference (LoG) **2025**

**Journal Reviewer**

- ACM Transactions on Knowledge Discovery from Data (TKDD)
- IEEE Transactions on Big Data (TBD)

**Honors** \_\_\_\_\_

- ICML’26 Gold Reviewer Award **2026**
- NeurIPS’25 Scholar Award **2025**
- VEF Fellow **2021**
- Vingroup Scholarship Recipient **2021**
- ACM-ICPC South Central Bronze Medal **2019**